

Executive overview

Construction Digital Transformation | Inventory + Damage Claims

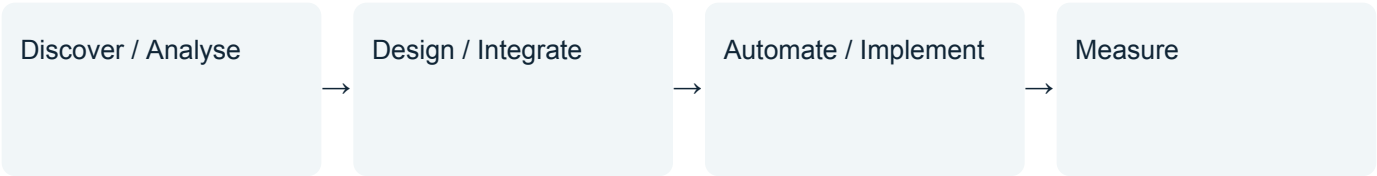
BuildFlow Group tracks reusable equipment and damage incidents through spreadsheets, paper and disconnected messages. Location and custody become unclear, evidence lacks consistent ownership, and Finance manually reconciles repairs. The case investigates those handoffs and designs an integrated operational workflow with a controlled implementation plan.

Scope: reusable inventory and damage claims. Role: independent analysis, process design, solution architecture and controlled implementation planning.

Journey: Discover → Analyse → Design → Integrate → Automate → Implement → Measure.

Status: design documented, prototype tested, enterprise validation pending.

Transformation journey



Designed	Demonstrated	Pending
Microsoft target + delivery plans	Tested local / browser prototype	Real workshops, tenant, ERP, UAT, benefits

Source: 01-discovery/business-context.md

Discovery and business analysis

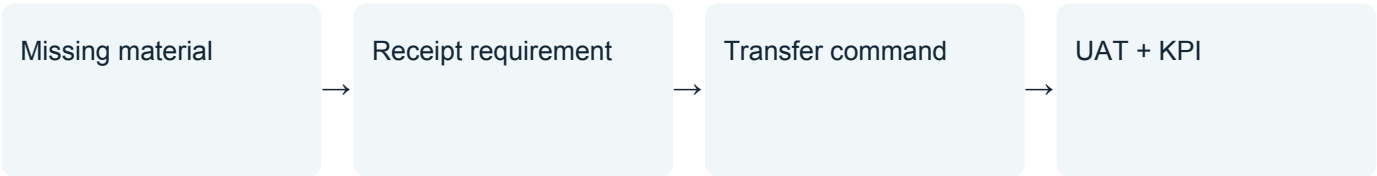
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Warehouse, site employees, project managers, Finance, Procurement, Operations and IT bring different constraints. The fictional workshop uses a walkthrough, Five Whys and MoSCoW. “Material goes missing” becomes an accountable transfer requirement.

PB-01 / BR-01 → FR-04 / AC-04 → transfer receipt component → UAT-INV-004 → KPI-02. The cause remains a hypothesis until sampled handoffs confirm it.

The analyst facilitates. Operations decides custody policy; Finance owns spend controls.

From complaint to accountable receipt



Stakeholder	Constraint	Decision owner
Warehouse / Sites	Fast work; uneven connectivity	Operations
Finance	Spend control; invoice matching	Finance
IT	Legacy ERP; supportability	IT Manager

Source: 01-discovery/workshop-notes-example.md

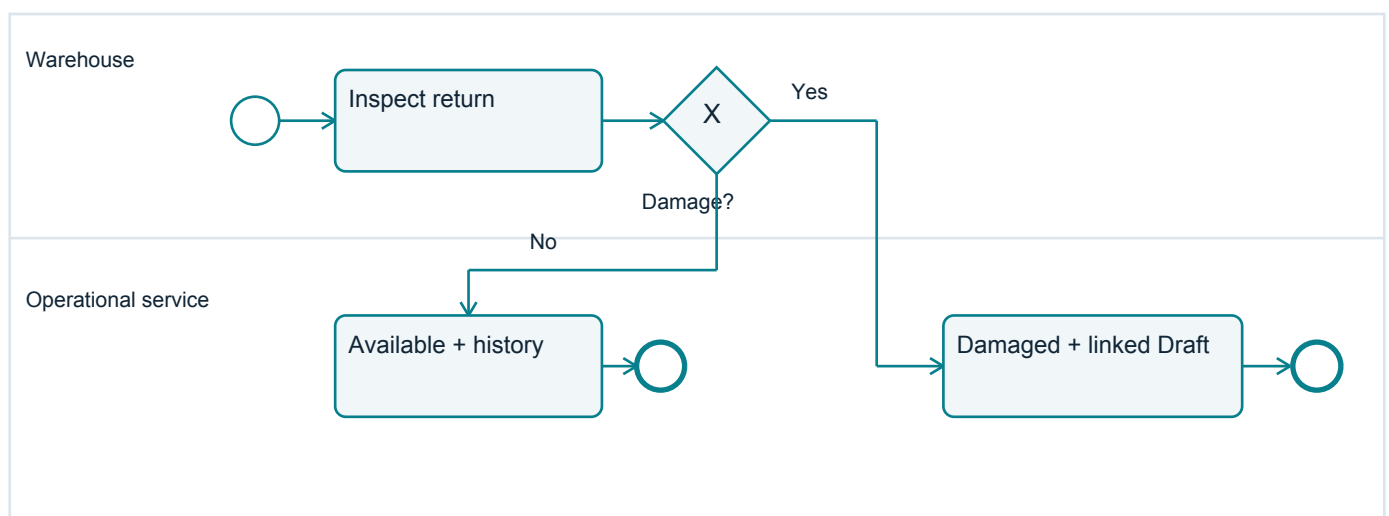
AS-IS and TO-BE

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AS-IS: spreadsheets and informal handoffs obscure current custody and return condition. TO-BE: a named custodian, confirmed transfer receipt and inspection gate establish operational truth.

The executive BPMN view focuses on the return: inspect condition, then either release Available or retain Damaged and create a linked Draft claim. Safety release is independent of claim closure. Offline stock movement remains unconfirmed until reviewed online.

Return inspection — executive BPMN view



Derived from the full BPMN source. Full model contains events, lanes and exception paths. Claims do not establish employee liability.

Source: 03-processes/bpmn/inventory-return.bpmn

Solution architecture

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Target: Power Apps, Dataverse command rules, Power Automate, SharePoint and Entra ID. A dedicated adapter sends approved repair commitments to ERP and returns posted actuals.

ERP owns projects, suppliers, cost centres, capital values and invoice accounting. The operational solution owns custody, claims and approval history. Atomic writes protect state/history/outbox; asynchronous delivery tolerates ERP maintenance.

The tested local API and public browser simulation are implementation evidence of process rules, not a deployed Microsoft tenant.



Boundary	Control
Identity / evidence	Entra roles; SharePoint restricted evidence
Operational command	State, audit and outbox change atomically
Financial integration	Stable reference; deduplication; reconciliation
Monitoring	Correlation IDs; retry queue; dead-letter ownership

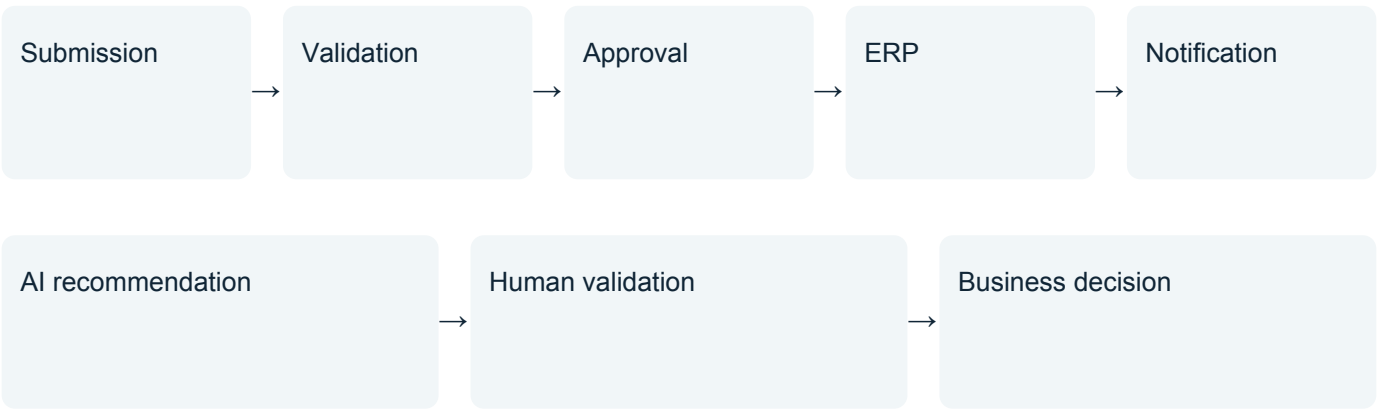
Source: 04-solution-design/architecture.md

Automation and AI

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Claim submission produces validation, ownership, a durable approval task and notification. Manager approval is required; above EUR 5,000 Finance must also approve the same revision. Changed amount/evidence invalidates prior approvals.

Optional AI structures facts and flags missing information. Human review precedes any business decision. AI cannot determine liability, sanction employees, approve spend or post invoices. Live demo uses deterministic suggestions; provider, privacy and quality evaluation remain open.



Power Automate blueprints are design artefacts. The browser demonstrates role-based approval rules and mock ERP delivery; it does not execute cloud flows.

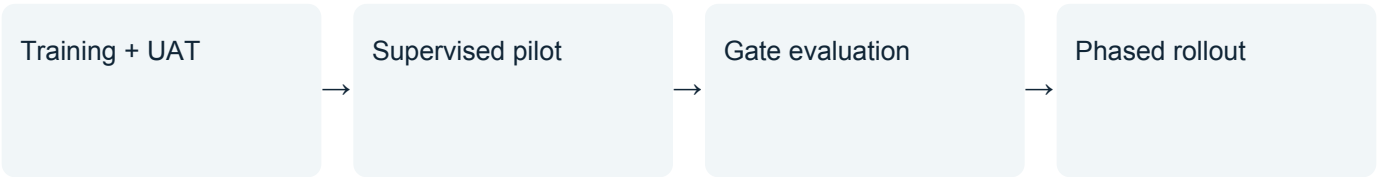
Source: 06-automation/power-automate/02-high-value.md

Controlled implementation

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Pilot one warehouse and two representative sites. Gate entry on unique labelled assets, role mapping, ERP contract checks and task training. Run 24 prepared UAT scenarios including concurrency and exception cases.

Migration stages and quarantines questionable data. Rollback preserves accepted movements and reconciles ERP references. Champions, observed task practice, feedback and ten working days of hypercare support adoption. Business, Warehouse, Finance and IT sign their own gates.



Gate	Evidence required
Business readiness	Task success + exception acceptance
IT readiness	Identity, recovery and monitoring verified
Finance readiness	Commitment / invoice reconciliation
Adoption	Champions, observed training, feedback
Support	Ten working days hypercare; named owners

Source: 10-implementation/rollout-plan.md

What this demonstrates

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The case links an operational problem to requirements, process, technical boundaries and controlled delivery. Evidence shows analysis and design reasoning, plus tested prototype rules. It does not claim real stakeholder facilitation or client production results.

Illustrative targets: inventory accuracy 92% to at least 98%; claim resolution 18 to at most 10 business days; same-shift capture 55% to at least 95%. Establish the real baseline before judging value. Cash avoidance and released capacity remain separate.

Role requirement	Evidence
Business ↔ IT	End-to-end traceability
BPMN	Process models + exceptions
APIs / ERP	Contracts + ownership / reconciliation
Power Automate	Durable approval blueprints
AI	Opportunity assessment + human review
Implementation	UAT + rollout + change plan

Measure	Illustrative baseline → target
Inventory accuracy	92% → ≥98%
Claim resolution time	18 business days → ≤10 business days
Same-shift capture	55% → ≥95%

Source: 13-kpis/target-kpis.md